

Lesson 27: Correlation & Linear Regression — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Describe $r = 0.95$.
2. Describe $r = -0.9$.
3. Describe $r = 0.1$.
4. Does correlation prove causation?
5. $y = 2x + 5$. Predict y at $x = 4$.
6. Range of values for r ?

Section 2 — Core practice

7. Describe $r = -0.82$.
8. Describe $r = 0.55$.
9. $y = 3x + 1$. Predict y at $x = 6$.
10. $y = -2x + 10$. Predict y at $x = 3$.
11. $y = 6.4x + 21$. Predict y at $x = 5$.

12. Is predicting far outside the data range reliable?

Section 3 — Challenge & exam-style

13. For 10 students, $r = 0.86$ and $y = 6.4x + 21$ (x = hours, y = score). Describe the correlation and estimate the score for 4 hours.

14. A regression line is $y = -1.5x + 40$. Interpret the gradient.

15. Why can a strong correlation between ice-cream sales and drownings be misleading?

16. Given $y = 4x + 2$ with data for x from 1 to 10, is a prediction at $x = 25$ reliable? Explain.